

BIJAN MAZAHERI

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I am interested in the information theory of causality, experimental design, and artificial intelligence.

EDUCATION

California Institute of Technology - Pasadena, CA <i>Doctor of Philosophy</i> Computing and Mathematical Sciences, GPA: 3.9/4.0 Thesis Title: Combining Sources and Leveraging Contexts Advised by Prof. Leonard Schulman and Prof. Jehoshua (Shuki) Bruck	<i>Oct 2017 – Aug 2023</i>
University of Cambridge - Cambridge, UK Classes in the Mathematical Tripos Parts IA, IB, and II. Supported by a Herchel Smith Fellowship.	<i>Oct 2016 – Jun 2017</i>
Williams College - Williamstown, MA <i>Bachelor of Arts</i> Physics and Computer Science, GPA: 3.92/4.00 Highest Honors (Physics), Phi Beta Kappa, Sigma Xi, Magna Cum Laude Thesis Title: RNA Macrostates and Macrokinetics	<i>Sep 2012 – Jun 2016</i>

ACADEMIC APPOINTMENTS

Thayer School of Engineering, Dartmouth College - Hanover, NH <i>Assistant Professor of Engineering</i> Affiliate - Quantitative and Biological Sciences Affiliate - Computational Science and Modeling Program	<i>Jan 2025 – Present</i>
Broad Institute of MIT and Harvard - Cambridge, MA <i>Eric and Wendy Schmidt Postdoctoral Associate</i> <i>Visiting Research Scientist</i> Primarily advised by Prof. Caroline Uhler.	<i>Oct 2023 – Dec 2024</i> <i>Jan 2025 – Present</i>

INDUSTRY EXPERIENCE

Operand – San Francisco, CA <i>Founding Research Advisor</i>	<i>Apr 2026 – Present</i>
Amazon Research Causality Lab - Tübingen, Germany <i>Intern</i> Worked with Dr. Michaela Hardt, Dr. Atalanti Mastakouri, and Dr. Dominik Janzing	<i>Oct 2022 – Feb 2023</i>
BioDiscovery - El Segundo, CA <i>Intern</i> Worked with Dr. Soheil Shams.	<i>Jun 2017 – Sep 2017</i>
IBM T.J. Watson Research Center - Yorktown Heights, NY <i>Intern</i> Worked with Dr. Victor Kravets (mentor) and Dr. Andrew Sullivan (manager).	<i>Jun 2016 – Sep 2016</i>

AWARDS AND GRANTS

Dartmouth Initiative for Middle East Exchange (DIMEX) “AISTATS 2026 and Resulting Collaborations.” \$22,000.	<i>Awarded Fall 2025</i>
DARPA Advanced Research Concepts - COMPASS (Lead PI) “Detecting Data Manipulation via Causal Information Geometry.” \$248,000.	<i>Awarded Summer 2025</i>
Eric and Wendy Schmidt Postdoctoral Fellowship 1–3 years of research at the Broad Institute of MIT and Harvard. \$225,000 approx.	<i>Awarded Summer 2023</i>
Amazon AI4Science Research Fellowship “Expert Graphs and Source Identification.” \$25,000.	<i>Awarded Spring 2022</i>
National Science Foundation Graduate Research Fellowship 3-year Ph.D. fellowship. \$138,000 approx.	<i>Awarded Spring 2019</i>
Herchel Smith Fellowship Funding for 1–3 years of study at the University of Cambridge. \$100,000 approx.	<i>Awarded Spring 2016</i>

TEACHING

Dartmouth Engineering Discrete and Probabilistic Systems Principles of Causality	<i>Summer 2025, Fall 2026</i> <i>Spring 2025, 2026</i>
Teaching Assistant Markov Chain Monte Carlo (Caltech) Discrete Mathematics (Williams) Electricity and Magnetism (Williams) Introduction to Mechanics (Williams)	<i>Spring 2022</i> <i>Fall 2015</i> <i>Spring 2015</i> <i>Fall 2013</i>

PUBLICATIONS

- Sophia Xiao and **Bijan Mazaheri**. Causal Discovery from Unseen Environments. Accepted to NeurIPS, 2026.
- Bijan Mazaheri**, Jiaqi Zhang, and Caroline Uhler. [Relaxing Faithfulness for Intervention-Only Causal Discovery](#). In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2026.
- Bijan Mazaheri**, Jiaqi Zhang, and Caroline Uhler. [Meta-Dependence in Conditional Independence Testing](#). In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2026.
- Bijan Mazaheri**, Spencer Gordon, Yuval Rabani, and Leonard J. Schulman. [Causal Discovery under Latent Class Confounding](#). In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2026.
- Bijan Mazaheri**, Chandler Squires, and Caroline Uhler. [Synthetic Potential Outcomes and Causal Mixture Identifiability](#). In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.
- Bijan Mazaheri**, Siddharth Jain, Matthew Cook, and Jehoshua Bruck. [Omitted Labels Induce Nontransitive Paradoxes in Causality](#). In *Conference on Causal Learning and Reasoning (CLeAR)*, 2025.
- Spencer Gordon, Erik Jahn, **Bijan Mazaheri***, Yuval Rabani, and Leonard J. Schulman. [Identification of Mixtures of Discrete Product Distributions in Near-Optimal Sample and Time Complexity](#). In *Conference on Learning Theory (COLT)*, 2024.

Bijan Mazaheri, Atalanti Mastakouri, Dominik Janzing, and Michaela Hardt. [Causal Information Splitting: Engineering Proxy Features for Robustness to Distribution Shifts](#). In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2023.

Spencer Gordon, **Bijan Mazaheri***, Yuval Rabani, and Leonard J. Schulman. [Causal Inference Despite Limited Global Confounding via Mixture Models](#). In *Conference on Causal Learning and Reasoning (CLEaR)*, 2023.

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. [Glioblastoma Signature in the DNA of Blood-Derived Cells](#). *PLOS ONE*, 16(9): e0256831, 2021.

Bijan Mazaheri, Siddharth Jain, and Jehoshua Bruck. [Expert Graphs: Synthesizing New Expertise via Collaboration](#). In *IEEE International Symposium on Information Theory (ISIT)*, 2021.

Spencer Gordon, **Bijan Mazaheri***, Yuval Rabani, and Leonard J. Schulman. [Source Identification for Mixtures of Product Distributions](#). In *Conference on Learning Theory (COLT)*, 2021.

Bijan Mazaheri, Siddharth Jain, and Jehoshua Bruck. [Robust Correction of Sampling Bias Using Cumulative Distribution Functions](#). In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

* Authorship order is alphabetical.

PREPRINTS

Hamza Virk, **Bijan Mazaheri**, and Yihren Wu. [The Spectral Structure of Latent Treatment Effects](#). 2026.

Zou Yang, Sophia Xiao, and **Bijan Mazaheri**. [Masking Causality and Conditional Dependence](#). Under review, 2026.

Liyuan Xu and **Bijan Mazaheri**. [Estimating Aleatoric Uncertainty in the Causal Treatment Effect](#). Under review, 2026.

Mateusz Gajewski, Sophia Xiao, and **Bijan Mazaheri**. [Data Augmentation via Causal Residual Bootstrapping](#). Under review, 2026.

Spencer Gordon, **Bijan Mazaheri***, Yuval Rabani, and Leonard J. Schulman. [The Sparse Hausdorff Moment Problem, with Application to Topic Models](#). *arXiv:2007.08101*, 2020.

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. [Cancer Classification from Healthy DNA using Machine Learning](#). *bioRxiv:517839*, 2019.

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. [Short Tandem Repeats Information in TCGA is Statistically Biased by Amplification](#). *bioRxiv:518878*, 2019.

* Authorship order is alphabetical.

PATENTS

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. [Mutation Profile and Related Labeled Genomic Components, Methods and Systems](#). US Patent App. 20190392951A1, 2019.

INVITED TALKS

Dartmouth Homecoming Back to Class Lecture “Ranking Runners who Never Raced”	<i>Oct 2026</i>
Air Force Research Laboratory QuEST Meeting “Modern Causal Inference and Defense”	<i>Jun 2026</i>
CauScale Spotlight @ AISTATS “Masked Unfairness: Hiding Causality in Zero ATE”	<i>May 2026</i>
Tsinghua Sanya International Mathematics Forum “Distribution-Level Techniques for Heterogeneous Causal Relationships”	<i>Jan 2026</i>
Biomedical Data Science Seminar Series, Dartmouth Medical School “Causal Underpinnings of Information Synthesis”	<i>Oct 2025</i>
Williams College Statistics Colloquium “Causal Underpinnings of Information Synthesis”	<i>Oct 2025</i>
Ops/MS Brown Bag Seminar, Tuck School of Business “Synthetic Potential Outcomes and Causal Mixture Identifiability”	<i>Feb 2025</i>
Boston University Machine Learning Symposium “Synthetic Potential Outcomes and Causal Mixture Identifiability”	<i>Nov 2024</i>
Stanford Online Causal Inference Seminar “Synthetic Potential Outcomes and the Hierarchy of Causal Identifiability”	<i>Oct 2024</i>
Jones Seminar, Thayer School of Engineering at Dartmouth “Latency and Heterogeneity in Data and What to Do About It”	<i>May 2024</i>
Simons Institute for the Theory of Computing “Causal Discovery under Limited Global Confounding”	<i>May 2023</i>

RESEARCH MENTORSHIP

PhD Advisees

Hamza Virk (Thayer School of Engineering, Dartmouth)
Zou Yang (Thayer School of Engineering, Dartmouth)
Sophia Xiao (Thayer School of Engineering, Dartmouth)

Graduate Students

Mateusz Gajewski (PhD Student, Poznań University of Technology)
Ryan Montgomery (MBA Student, Tuck School of Business, Dartmouth)

Undergraduate Students

Haley Lin (Williams College)
Rafael Castro (California Institute of Technology SURF)
Youmi Ji (Dartmouth College URAD)
Keion Grieve (Dartmouth College URAD)
Caroline Krantz (Dartmouth College URAD)
Muhammad Ahmad (Dartmouth College FYREE)
Hailey King (Dartmouth College)
Ruchita Nair (Dartmouth College)
Benjamin Cavanagh (Dartmouth College)

Sreshth Tiwari (Dartmouth College URAD)
Connor Kilkenny (Dartmouth College URAD)
Titus Johnson (Dartmouth College URAD)
Tamier Baoyin (B.A. Mt. Holyoke College, B.E. Dartmouth College)
Leila Salken (Williams College)
Anika Roy (B.S. IIIT Hyderabad, Khorana Scholar, Heidelberg Future Leaders Award)
Beshr Bouli (MIT UROP)

Data Science Projects

I have supported over 30 projects with undergraduate students using data on my website, and have advised some of these students in applying to graduate school.

WORKSHOPS

CauScale @ AISTATS “Masked Unfairness: Hiding Causality in Zero ATE”	<i>May 2026</i>
Tsinghua Sanya International Mathematics Forum “Causality and Machine Learning”	<i>Jan 2026</i>
CauScien Workshop at NeurIPS Attendee.	<i>Dec 2025</i>
Scaling Up Intervention Models at ICML “Faithfulness and Intervention-Only Causal Discovery”	<i>Jul 2025</i>
Simons Institute for the Theory of Computing “Causality”	<i>Spring 2022</i>

SERVICE

Conferences

CLear: Reviewer (2024–2026), Program Chair (2026).
NeurIPS: Reviewer (2020–2025), Top Reviewer Award (2025).
UAI: Reviewer (2026).
AISTATS: Reviewer (2023–2026).
ICML: Reviewer (2025).

Journals

Nature Machine Intelligence: Reviewer (2024)
The American Statistician: Reviewer (2024)
Electronic Journal of Statistics: Reviewer (2025)

Committees

Dartmouth Engineering 3-2 and 2-1-1-1 Programs
Curriculum, Dartmouth Computational Science and Modeling Program

Other Service

Faculty Advisor, Dartmouth Cross Country and Track and Field Teams
Volunteer Assistant Coach, Caltech Cross Country and Track and Field Teams
Sports statistics outreach for Cross Country and a correspondent at D3 Glory Days

PROJECTS

LACCTiC

Sep 2021 – Present

I maintain a website for collegiate cross country with 10,000 regular users that applies concepts from batch-effect correction to ranking performances on differing terrain. The backend runs on Python and Django and the frontend uses React, and the database is hosted on AWS. I have helped advise over 30 student projects using this data.